

Europeans Lead

The advent of the 21st century saw many developments in the lighter side of general aviation, including the introduction of diesel and electric engines, ballistic recovery systems (BRS), and the recognition of an entirely new class of airplane, the FAA's Light Sport Aircraft (LSA). However, the biggest changes were in avionics, with even small two-seat aircraft being fitted with advanced autopilots, traffic collision avoidance systems (TCAS), and synthetic vision—equipment that would not look out of place in an Airbus or a big Boeing.



△ Cirrus SR22 2000

Origin USA

Engine 310 hp Continental Motors IO-550 turbocharged air-cooled flat-6

Top speed 231 mph (372 km/h)

Derived from the Cirrus SR20, the 22 is a significantly more powerful version because it is fitted with a turbocharged engine. It has the same ballistic recovery systems as the SR20, and also features a very powerful avionics suite.

◁ Diamond DA42 Twin Star 2002

Origin Austria

Engine 2 x 165 hp Austro turbocharged liquid-cooled diesel in-line 4-cylinder

Top speed 222 mph (356 km/h)

Made primarily from composite material, the DA42 has very impressive performance, particularly for range and endurance, because of its extremely efficient diesel engines and advanced aerodynamics.

▷ Alpi Pioneer 300 2006

Origin Italy

Engine 100 hp Rotax 912 ULS liquid-cooled flat-4

Top speed 168 mph (270 km/h)

Looking like a scaled-down version of a Siai Marchetti SF-260, Alpi Aviation's sleek, kit-built two-seat retractable boasts an impressive cruise speed for an aircraft that only has a 100 hp engine.



◁ Alpi Pioneer 400 2010

Origin Italy

Engine 115 hp Rotax 914 turbocharged liquid-cooled flat-4

Top speed 184 mph (296 km/h)

Derived from the successful Pioneer 300 two-seater, the retractable tricycle undercarriage and turbocharged engine ensure high cruise speeds. Although a very modern design, the Pioneer 400 is unusual for a 21st-century aircraft because it is constructed primarily from wood.



Light Sport Aircraft

The introduction of the new category of LSA rejuvenated sport aviation in the US. These aircraft could be flown on a new type of license—the Sport Pilot Certificate—which has less strict medical requirements. These aircraft have an unpressurized cabin, fixed landing gear, and no more than two seats. They are limited to a weight of 1,321 lb (599 kg) and a stall speed of no more than 52 mph (83 km/h).



△ American Legend Cub 2004

Origin USA

Engine 100 hp Continental O-200 air-cooled flat-4

Top speed 108 mph (174 km/h)

The introduction of the LSA class by the FAA saw many new machines come to market, including some "retro" designs. These include this Legend Cub, which is based on the classic Piper J3 Cub, but built using modern materials and methods, and fitted with advanced instruments and avionics.